# Non-Linearity, Tree-Based Classification & SVM

Course Project 4 · CSCI-6767 Applied Machine Learning & Data Analytics

Spring 2026

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# Agenda


**1** **Problem Statement & Motivation**

Why non-linear? Why two datasets?

**2** **Dataset 1 — Seoul Bike Sharing Demand**

Regression task — EDA & non-linear patterns

**3** **Task 1 — Non-Linear Models**

Polynomial · Step Functions · Splines · LOESS · GAMs

**4** **Dataset 2 — KDD Cup 1999 Network Intrusion**

Classification task — 494k connections

**5** **Task 2 — Tree-Based Methods**

Bagging · Random Forests · Gradient Boosting

**6** **Task 3 — Support Vector Machines**

Linear · RBF · Polynomial · Sigmoid kernels

**7** **Model Comparison & ROC Curves**

Full metrics tables and performance analysis

**8** **Key Findings & Conclusions**

What worked, what didn't, why it matters

# Problem Statement & Motivation

## What are we solving?

- Bike-sharing demand varies non-linearly with temperature, hour, and season
- Network intrusion patterns cannot be separated by a linear boundary
- Goal: apply and compare ML models across two real-world tasks

## Two-Dataset Strategy

- Regression dataset → Task 1 (non-linear models)
- Classification dataset → Tasks 2 & 3 (trees + SVM on same data)
- Enables direct apples-to-apples comparison within each task

### Required methods (from course syllabus)

Task 1 (30%): Polynomials · Step Functions · Splines · Local Regression (LOESS) · GAMs
Task 2 (30%): Bagging · Random Forests · Boosting (Gradient Boosting)
Task 3 (20%): SVM — Linear / RBF / Polynomial / Sigmoid kernels
Task 4 (20%): Model comparison, error analysis, and conclusions

Section 1

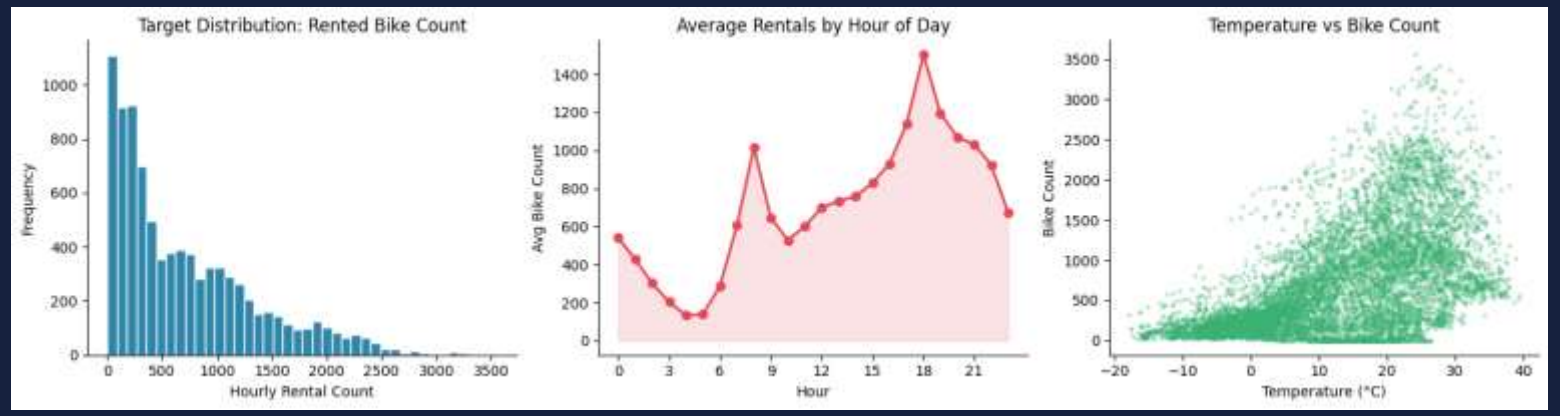
# Regression Task

Seoul Bike Sharing Demand — Non-Linear Models

# Dataset 1 — Seoul Bike Sharing Demand (Regression)

## Overview

- UCI ML Repository · auto-downloaded via ucimlrepo(id=560)
- 8,760 hourly records (Dec 2017 – Nov 2018, Seoul)
- Target: Rented Bike Count (0–3,556 bikes/hour)
- 14 features: temperature, humidity, hour, season, rainfall, …
- 0 missing values




## Why this dataset?

- Strong non-linear temperature effect (sweet spot 20–25°C)
- Bimodal hourly pattern (8am rush + 6pm rush) — linear models fail
- Seasonal structure ideal for GAMs and splines
- Clean, no preprocessing overhead → focus on modelling

# Task 1 — Non-Linear Models: Why & How

## Polynomial

$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2 + \ldots + \beta_d x^d$

Add degree-d terms. RidgeCV auto-selects λ after StandardScaler.

## Step Functions

$f(x) = \Sigma\, c_j \cdot 1(x \in [t_{j-1}, t_j))$

Bin continuous features; fit constant per bin. Piecewise constant.

## B-Splines

$f(x) = \Sigma\, \beta_j\, B_{j,}{}^{d}(x)$ *(n_knots=6, degree=3)*

Piecewise polynomials joined smoothly at knots. sklearn SplineTransformer.

## Nat. Cubic Splines

*cr(x, df=5) — linear outside boundary knots*

Cubic splines with linearity constraint in tails. patsy cr().

## LOESS

$\hat{y}(x_0)$ = *local WLS fit with tricube kernel*

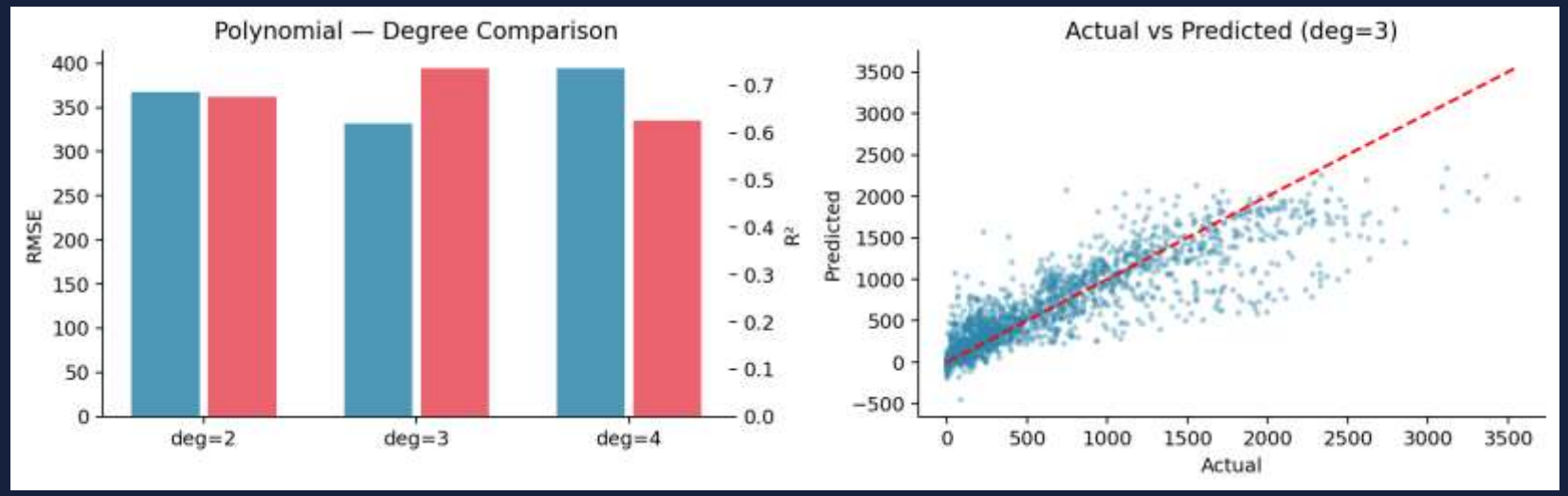
Non-parametric local smoother. Visualization tool — bandwidth = frac.

## GAM

$g(\mu) = \beta_0 + f_1(x_1) + f_2(x_2) + \ldots + f_p(x_p)$

Sum of smooth functions. pygam LinearGAM, λ tuned by grid search.

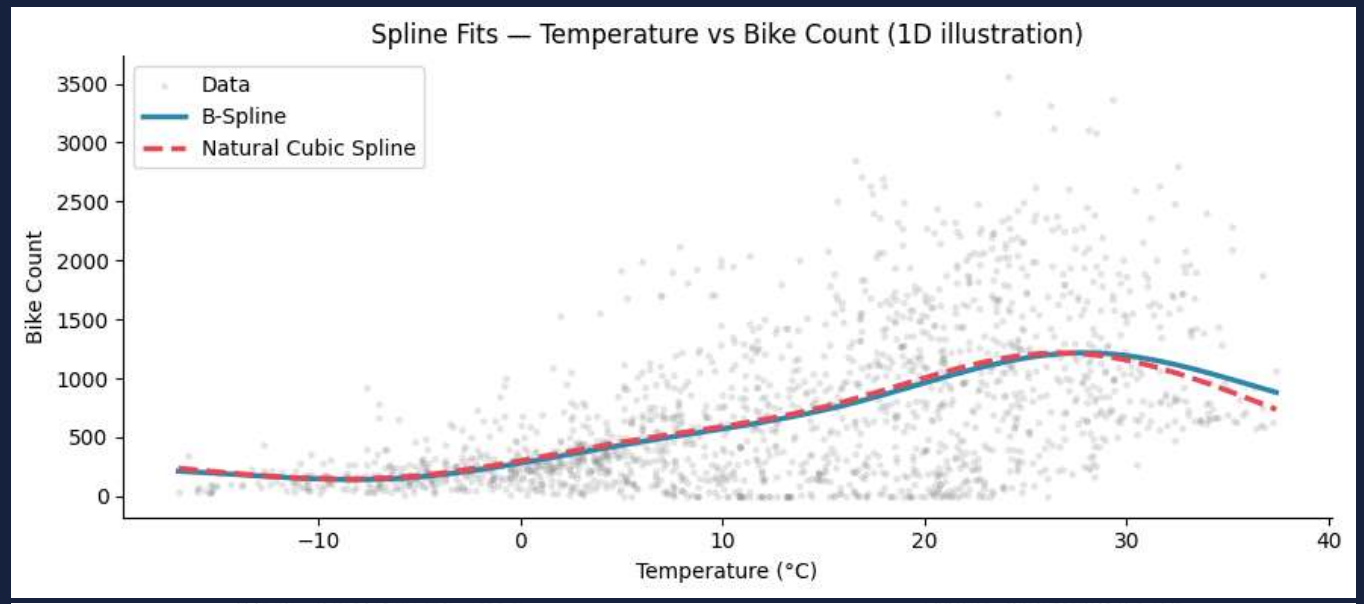
# Polynomial Regression & Step Functions — Results

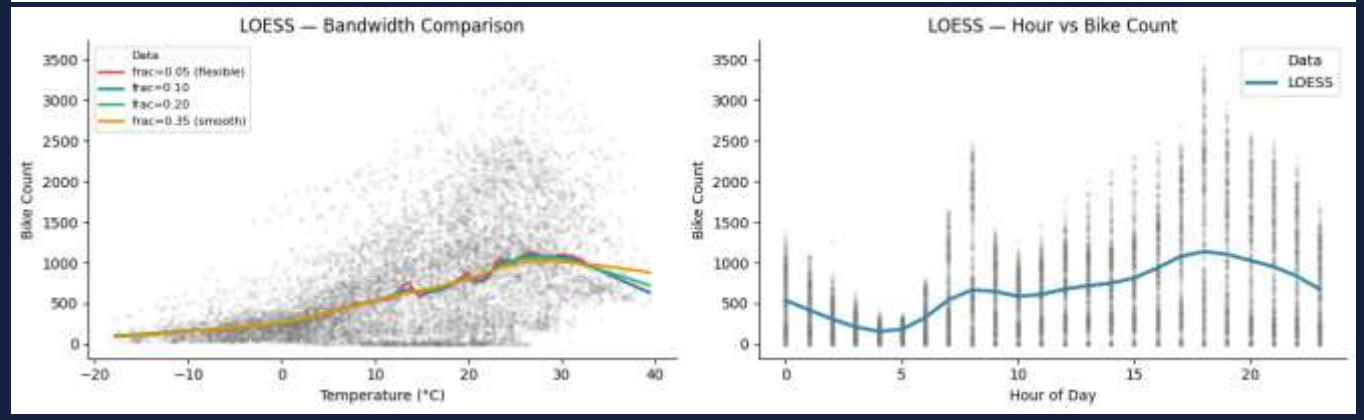



## Key results

- Best degree: 3 (RMSE=331, $R^2$=0.7366)
- StandardScaler essential before PolynomialFeatures
- Degree 4 (3,059 features) requires α≈10,000 via RidgeCV
- Step Functions: RMSE=401 $R^2$=0.615
- Step functions: piecewise constant → less smooth than splines

# Splines & LOESS Local Regression



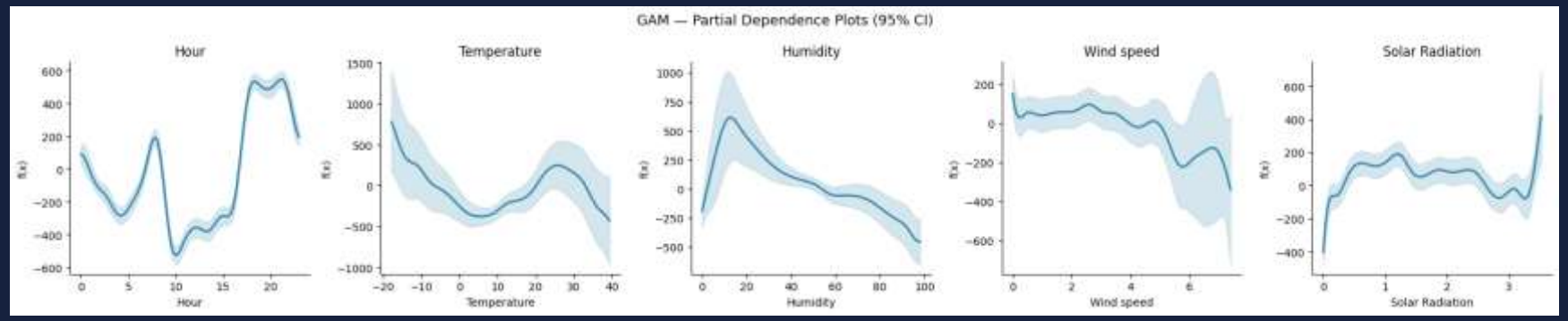

## B-Splines vs NCS

- B-Splines: sklearn SplineTransformer(n_knots=6, d=3)
- B-Splines RMSE=352 $R^2$=0.703
- NCS: patsy cr(x, df=5), linear outside boundary knots
- NCS RMSE=385 $R^2$=0.644
- Both outperform step functions — smooth, continuous fits

## LOESS

- Local non-parametric smoother — no global equation
- Bandwidth frac controls flexibility vs smoothness
- Used as diagnostic/visualization tool (not in RMSE table)
- Reveals non-monotonic temperature–demand relationship

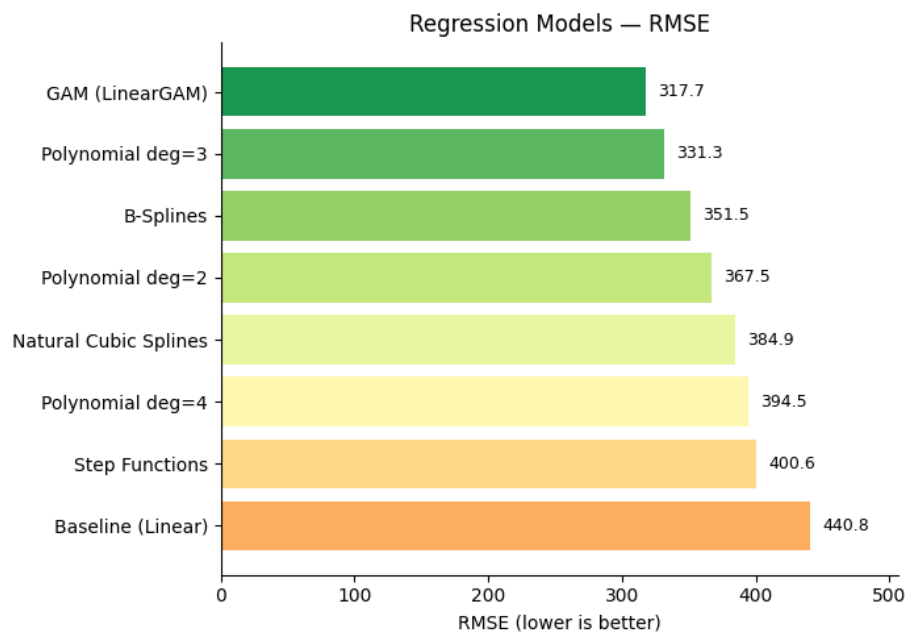
# Generalized Additive Models (GAMs) — Best Regression Model

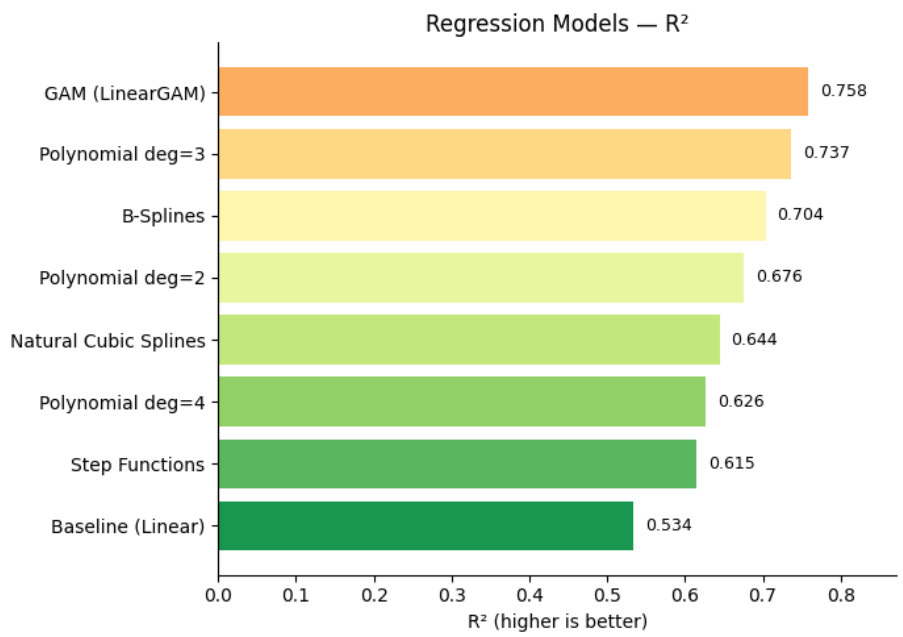



## GAM Results & Insights

- RMSE = 317.7 | $R^2$ = 0.7578 ← best regression model
- Formula: $\hat{y} = \beta_0 + f_1(\text{Hour}) + f_2(\text{Temp}) + f_3(\text{Humidity}) + \ldots$
- λ (smoothness penalty) tuned via grid search automatically
- Hour: clear bimodal peaks at 8am & 6pm — commuting pattern
- Temperature: non-linear bell curve peaking ~20°C
- Humidity: negative effect above 60%
- Partial dependence = marginal effect with all others at avg
- Key advantage: interpretable AND high-performance

# Regression Model Comparison — Full Results




## Rankings (by $R^2$)

- 1. GAM (LinearGAM) $R^2$=0.7578
- 2. Polynomial deg=3 $R^2$=0.7366
- 3. B-Splines $R^2$=0.7035
- 4. Polynomial deg=2 $R^2$=0.6759
- 5. Natural Cubic Splines $R^2$=0.6444
- 6. Baseline (Linear) $R^2$=0.5337
- All non-linear methods beat baseline — validates non-linearity
- GAM interpretable AND best-performing — recommended choice

Section 2 & 3

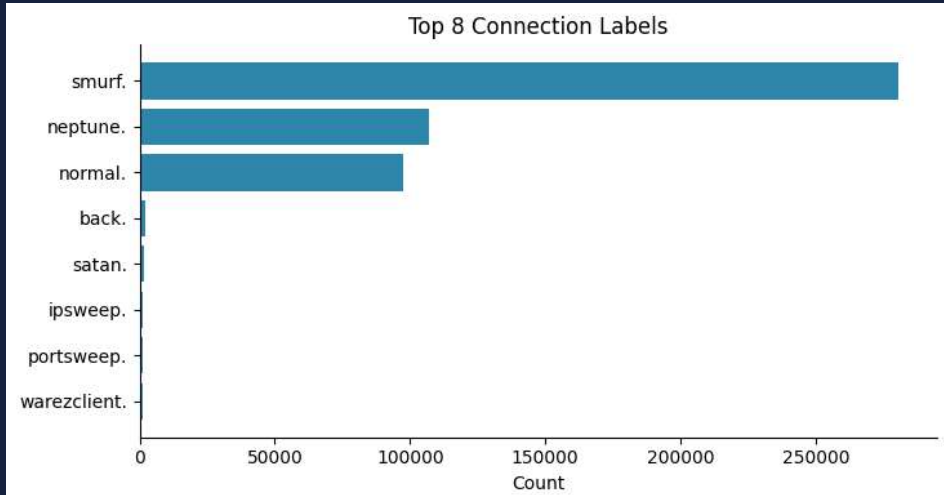
# Classification Task

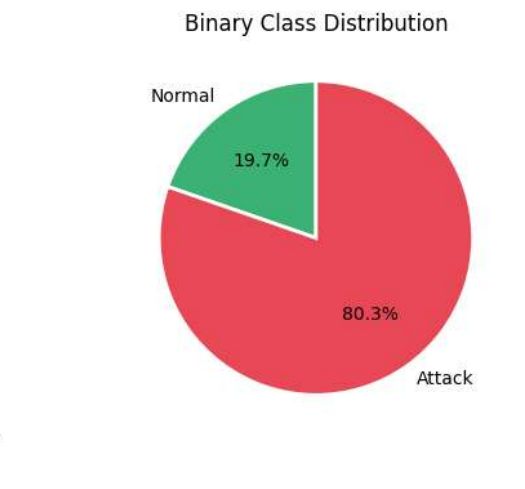
KDD Cup 1999 — Tree Methods & SVM

# Dataset 2 — KDD Cup 1999 Network Intrusion (Classification)

## Overview

- sklearn built-in · fetch_kddcup99(percent10=True) · no download needed
- 494,021 network connection records (10% sample of full dataset)
- Binary target: normal (0) vs attack (1)
- 41 features: src_bytes, dst_bytes, protocol, service, flags, …
- Attack rate: ~80% (imbalanced, dominated by smurf & neptune DoS)




## Attack types

- DoS — Denial of Service (smurf, neptune, back…)
- Probe — port scanning (satan, ipsweep, nmap…)
- R2L — Remote to Local (guess_passwd, ftp_write…)
- U2R — User to Root (buffer_overflow, rootkit…)

# Task 2 — Tree-Based Methods: How They Work

## Bagging

Bootstrap Aggregating

*Formula:* $\hat{f}(x) = (1/B)\ \Sigma\ f_\beta(x)$

- Train B trees on bootstrap samples (with replacement)
- Aggregate by majority vote (classification)
- Reduces variance; bias unchanged
- OOB (~37% left out per tree) = free validation
- Settings: B=100, base=DecisionTree

## Random Forest

Bagging + Random Feature Subsets

*At each split: sample* $m = \sqrt{p}$ *features*

- Decorrelates trees → better variance reduction than Bagging
- Key: dominant features don't control every split
- Feature importance via Gini impurity decrease
- OOB scoring; n_jobs=-1 for parallelism
- Settings: 100 trees, oob_score=True

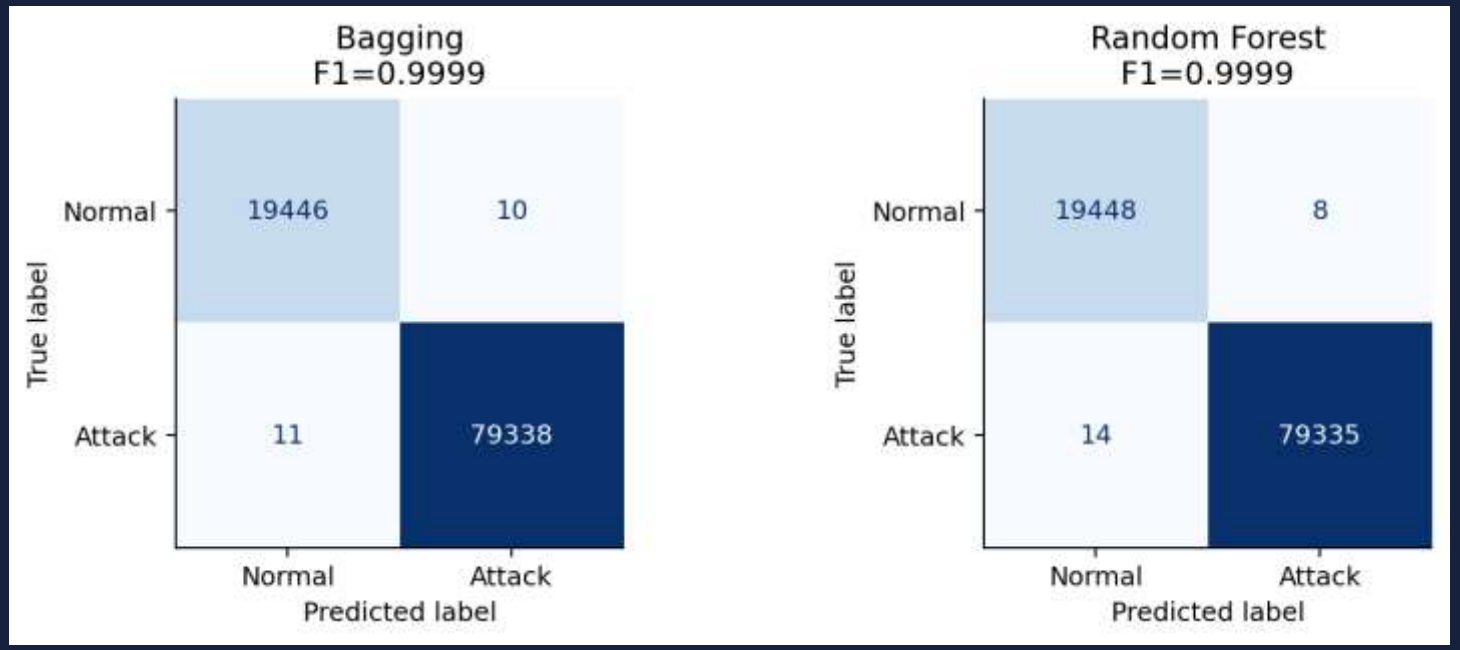
## Gradient Boosting

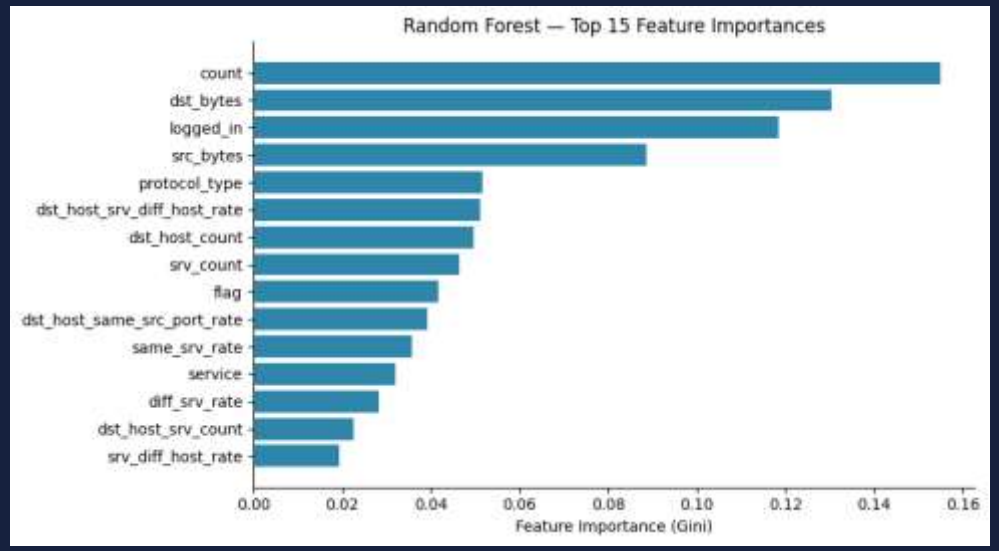
Sequential Error Correction

$F_m(x) = F_\{m-1\}(x) + v \cdot h_m(x)$

- Each tree fits the negative gradient (pseudo-residuals)
- Learning rate v=0.1 controls shrinkage
- Stochastic GB: subsample=0.8 per round
- Settings: 200 rounds, max_depth=4, lr=0.1
- Monitor train vs test deviance for early stopping

# Bagging & Random Forest — Results

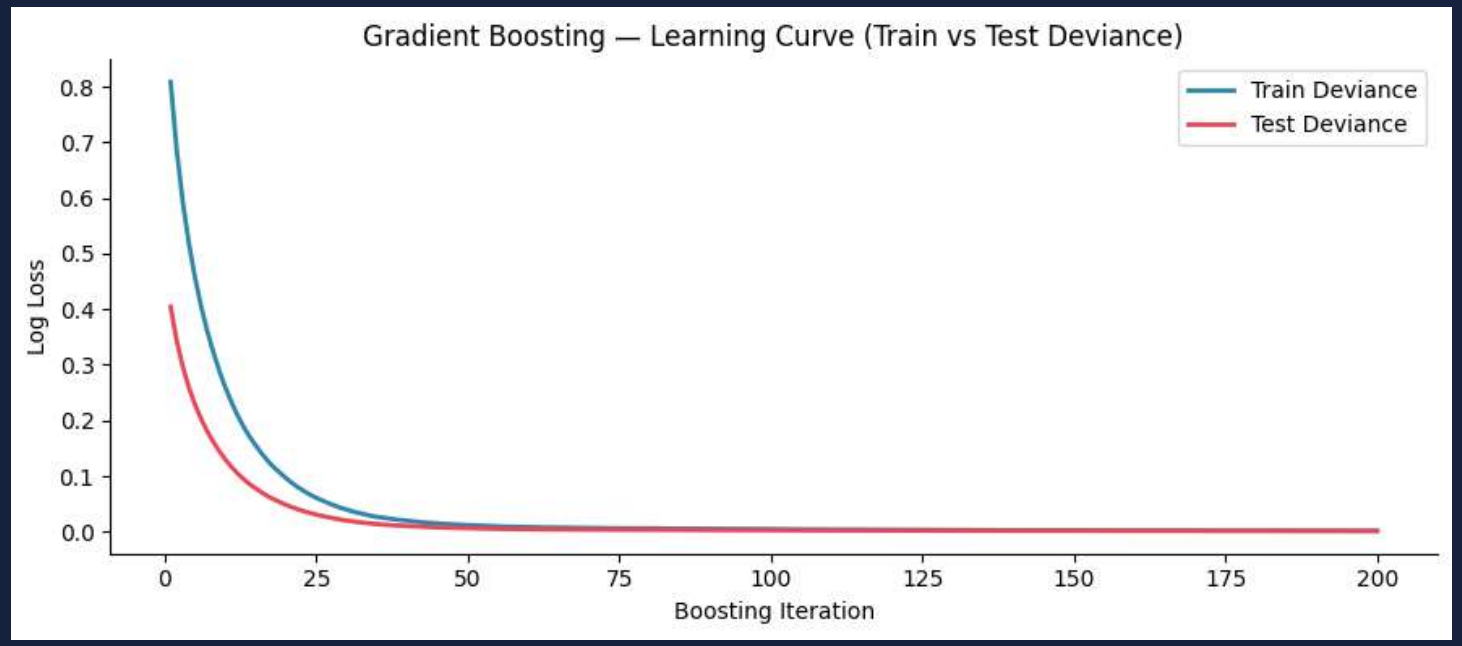


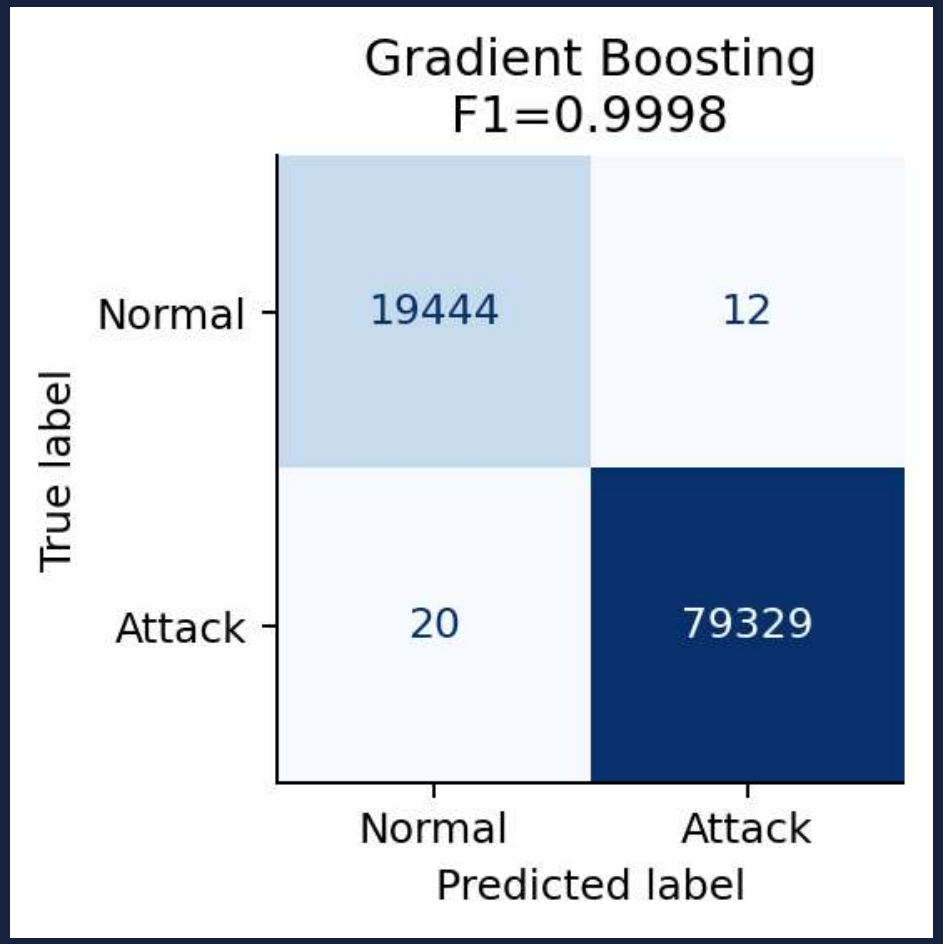




## Metrics

- Bagging: Acc=0.9998 F1=0.9999 AUC=1.0000 OOB=0.9998
- Random Forest: Acc=0.9998 F1=0.9999 AUC=1.0000 OOB=0.9998
- Both achieve near-perfect scores — KDD99 is highly separable
- RF feature importance: src_bytes, dst_bytes dominate
- Random feature sampling in RF slightly outperforms Bagging

# Gradient Boosting — Results & Learning Curve




## Observations

- Accuracy=0.9997  F1=0.9998
- AUC-ROC = 1.0000
- Train & test deviance converge → no overfitting
- 200 rounds with lr=0.1 provides stable generalisation
- Slightly lower than RF — sequential training adds noise
- Subsample=0.8 gives stochastic gradient boosting benefit

# Task 3 — Support Vector Machines: Kernel Functions

SVM finds the maximum-margin hyperplane. The kernel trick maps data to higher-dimensional space implicitly: $f(x) = \text{sign}(\Sigma\ \alpha_i\ y_i\ K(x_i, x) + b)$

## Linear

$K(x,z) = x^T z$

Finds linear separator in original space.
Best for linearly separable data.
C=1.0

Fast training · interpretable weights · may underfit non-linear data

## RBF (Gaussian)

$K(x,z) = \exp(-\gamma\|x-z\|^2)$

Maps to infinite-dimensional space.
Most general-purpose kernel.
C=10, γ=scale

Best performer here · handles non-linear boundaries · sensitive to C and γ

## Polynomial

$K(x,z) = (\gamma x^T z + 1)^d$

Captures polynomial interactions.
Degree d=3 used.
C=1.0

Good for data with polynomial structure · more params than RBF

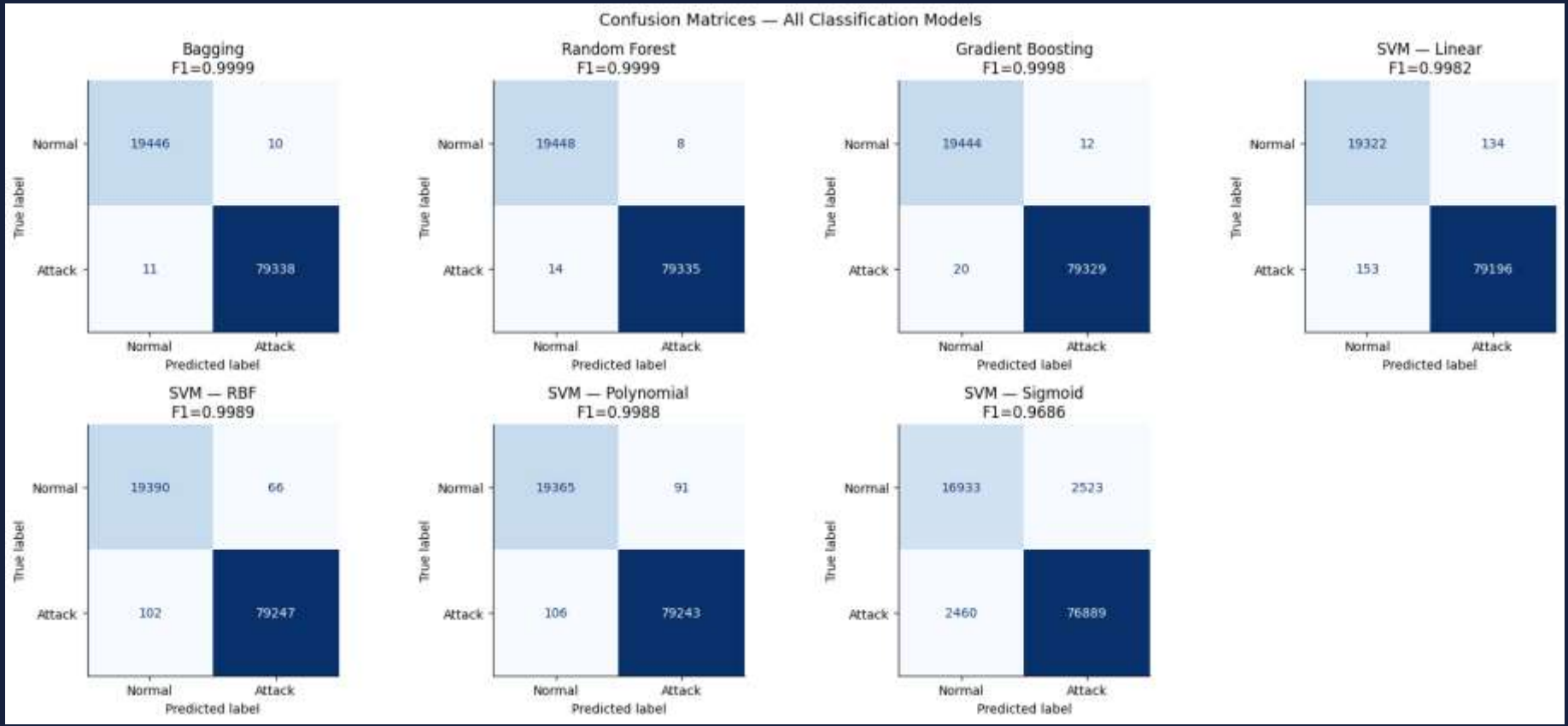
## Sigmoid

$K(x,z) = \tanh(\gamma x^T z + c)$

Neural network analogy.
Equivalent to 2-layer net.
C=1.0

Often underperforms · sensitive to scaling · not PD in general

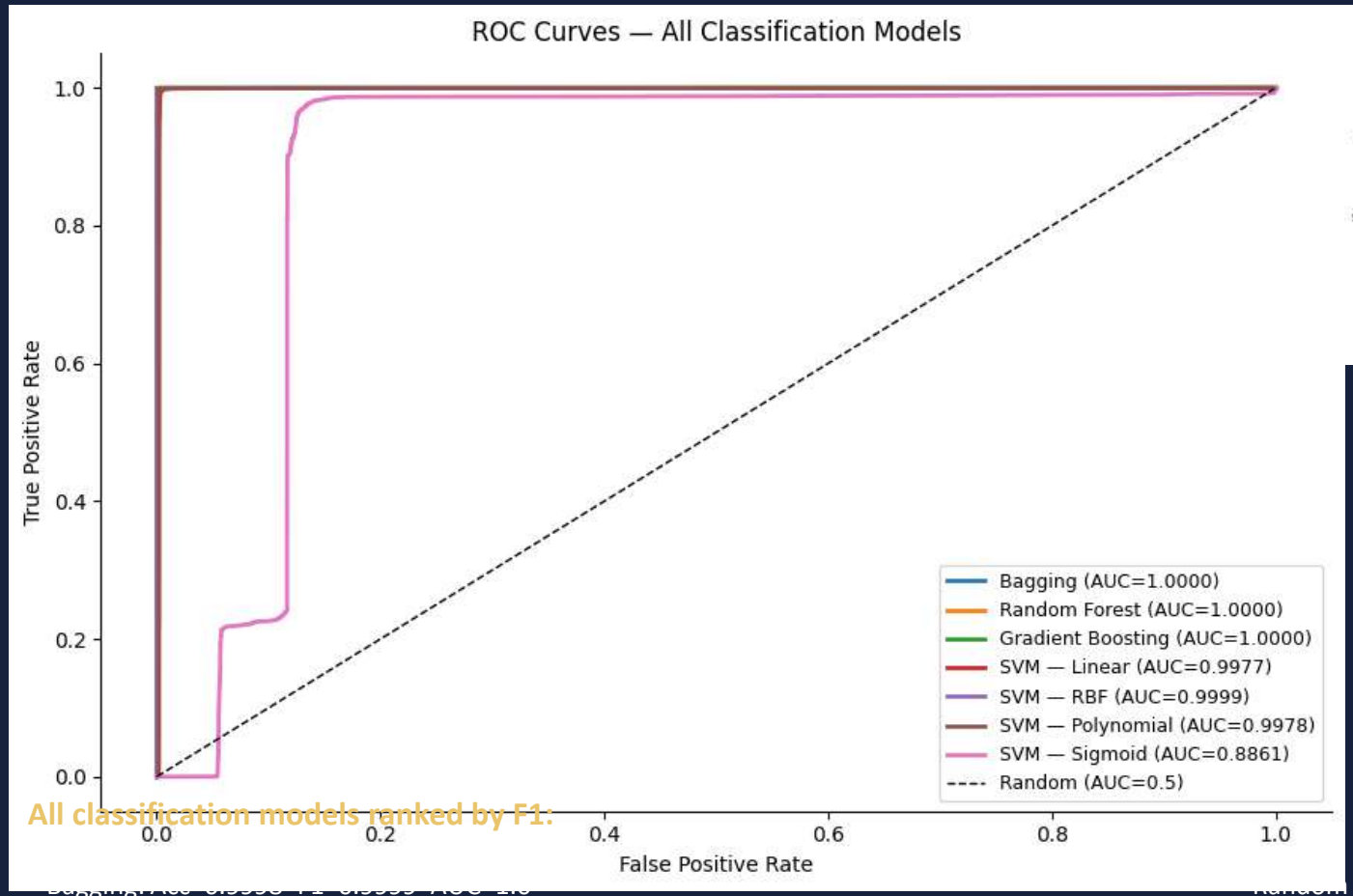
# SVM Kernel Comparison — Results

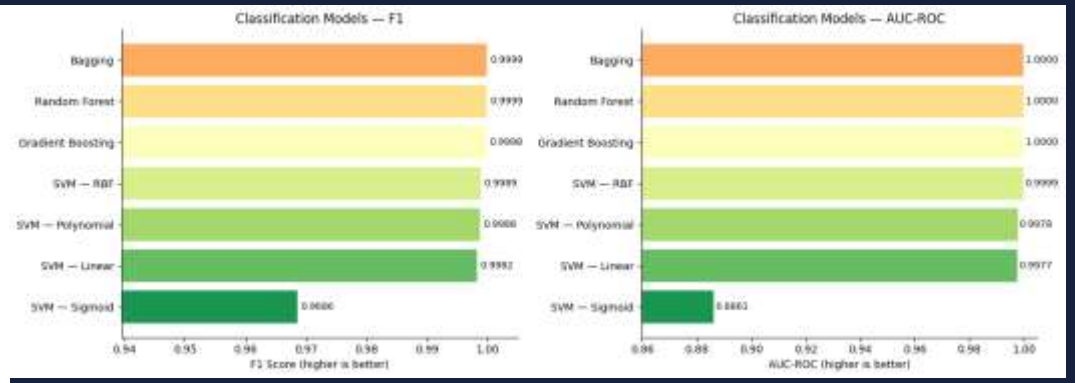



## Kernel Performance Summary

- Linear F1=0.9982 AUC=0.9977
- RBF F1=0.9989 AUC=0.9999
- Polynomial F1=0.9988 AUC=0.9978
- Sigmoid F1=0.9686 AUC=0.8861
- ⚠ All SVMs trained on 15k-sample subset ($O(n^2)$ constraint)
- RBF best — non-linear boundary in this feature space
- Sigmoid worst — unsuited for tabular data distributions

# Classification — Full Model Comparison & ROC Curves




All classification models ranked by F1:

Bagging: Acc=0.9998 F1=0.9999 AUC=1.0

Random Forest: Acc=0.9998 F1=0.9999 AUC=1.0

Gradient Boosting: Acc=0.9997 F1=0.9998 AUC=1.0

SVM — RBF: Acc=0.9983 F1=0.9989 AUC=0.9999

SVM — Polynomial: Acc=0.9980 F1=0.9988 AUC=0.9978

SVM — Linear: Acc=0.9971 F1=0.9982 AUC=0.9977

SVM — Sigmoid: Acc=0.9496 F1=0.9686 AUC=0.8861

Section 4

# Conclusions

Findings, Takeaways & Team Contributions

# Key Findings

## Regression — Seoul Bike Sharing

- Best model: GAM (LinearGAM) ($R^2$=0.7578, RMSE=318)
- All 5 non-linear methods beat linear baseline ($R^2$=0.53) — non-linearity confirmed
- Polynomial deg=3 competitive but requires StandardScaler + RidgeCV to be stable
- Splines give smooth fits; step functions produce discontinuous jumps
- LOESS reveals non-monotonic temperature effect — best diagnostic tool
- GAM: interpretable partial dependences + best RMSE — recommended model

## Classification — KDD Cup 1999

- Best model: Bagging (F1=0.9999, AUC=1.0)
- KDD99 is highly separable — all models achieve >94% accuracy
- RF and Bagging effectively tied — RF marginal advantage via decorrelation
- Gradient Boosting competitive — sequential learning captures residuals
- SVM RBF is best kernel — data boundary is non-linear in feature space
- SVM Sigmoid fails (AUC=0.886) — not suited for this tabular structure
- Tree ensembles dominate at scale; SVMs limited to ~50k training samples

# Conclusions & Takeaways

### → Non-linearity is critical

Every non-linear method outperforms linear regression on Seoul Bike data. Complex real-world patterns (hourly rhythms, temperature sweet-spots) cannot be captured by linear models regardless of regularisation.

### → GAMs: best of both worlds

LinearGAM achieves the lowest RMSE while remaining fully interpretable via partial dependence plots — the right tool when explainability matters alongside performance.

### → Ensemble methods dominate classification

Random Forests and Gradient Boosting set the performance ceiling at near-perfect F1. Parallelisable, scalable, and feature-importance aware — the recommended production choice.

### → SVM kernel selection matters

RBF kernel captures non-linear boundaries effectively (AUC≈1.00), while Sigmoid kernel fails on tabular data (AUC≈0.89). Kernel choice is a modelling decision — not a hyperparameter to grid-search blindly.

### → Two-dataset strategy pays off

Separating regression and classification tasks enables fair, within-task comparisons and avoids forcing models into mismatched settings.

# Team Contributions

## Gabil Gurbanov

- Task 1 — Non-Linear Models (all 5 methods)
- Polynomial Regression: scaling fix + RidgeCV degree selection
- Step Functions: binning strategy and one-hot encoding pipeline
- B-Splines (SplineTransformer) + Natural Cubic Splines (patsy cr())
- LOESS: bandwidth analysis and visualisation
- GAMs: pygam LinearGAM, grid search, partial dependence plots
- Regression EDA, preprocessing, model comparison & visualisations
- Streamlit app — regression sections

## Hamida Hagverdiyeva

- Task 2 — Tree-Based Methods (all 3 algorithms)
- Bagging: OOB scoring and bootstrap analysis
- Random Forest: feature importance, OOB evaluation
- Gradient Boosting: staged training, deviance tracking
- Task 3 — SVM (all 4 kernels): Linear, RBF, Polynomial, Sigmoid
- SVM kernel comparison, ROC curve analysis
- Classification dataset EDA, preprocessing, model comparison
- Streamlit app — classification sections

# Thank You

Questions & Discussion

Gabil Gurbanov · Hamida Hagverdiyeva
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